Question 21

Web API that follows a set of rules called REST, allows different systems to communicate over the internet using standard HTTP methods

R
E
S
T } Representational State Transfer

1) GET
→ Used to retrieve data from a server
2) POST
→ Used to sent or create new data on the server
3) PUT
→ Used to update existing data
4) DELETE
→ Used to remove data from the server.

Question 22

A lightweight format used to store & exchange data between systems. Represents data in a structured, readable text format using key-value pairs

Common Use

- Easy to read & write
- Easy for machines to parse & generate.
- Lightweight
- Language independent
- Perfect for web communication

Eg

```
"name": "Atleganc",
"Age": 22,
"City": Benon
```

Question 23

The skill of designing & refining inputs given to an AI system such as a large language model, to get the most accurate, useful & relevent output

It is important as the quality of the response depends on the prompt.

- Produces more accurate & relevant answers
- Reduces vague or incorrect response
- Help the AI follow specific formats or styles

Poor Prompt

- Tell me about marketing
- Write a cv

Improved

- Explain digital marketing in simple terms & include examples
- Write a proffessional cv for a recent IT graduate in SA with no experience

Question 24

A Programming tool used to apply a specific operation to every item in a collection and return a new collection with results

Function

- You give it a list of items
- You give it a function
- It applies that function to every item
- Returns a new list with transformed values

Example

```
Numbers [1,2,3,4,5]
# Use map to square each number
squared = list (map (lambda x: x * x, numbers))
print (squared)
```

Output

[1, 4, 9, 16, 25]

Question 25

The delivery of computing services, servers, databases, networking & software) over the internet istead of running them on a local computer or physical server

Cloud Service Models

1) Iaas (Infrastructure as a Service)
- Provides basic computing infrasture like virtual servers, storage & networking.

Example Amazon Web Services

2) Paas (Platform as a Service)
- Provides a platform for developers to build, test & deploy applications without managing infrastructure

Example Google Cloud App Engine

3) Saas (Software as a Service)
- Provides ready to use software applications over the internet.

Example Microsoft 365.

Question 26

VCS - a tool that helps track & manage changes to files overtime. Allows multiple people to work on a project & easily go back to earlier versions if something goes wrong

GIT - Allows developers & data professionals to track changes in code & files. Work collaboratively on the same project.

Widely used because
- keeps a full history of changes
- Supports teamwork without overwriting work
- Works well with platforms

1) Commit
a save point in a project
Records changes made to file at a specific time

2) Branch
a seperate version of your project where you can work on new features without affecting the main version.

Question 27

An interactive computing environment that allows you to write code, run it, & see results immediately in the same document.

Jupyter Notebooks are most commonly used with Python in machine learning, Also supports R & Julia through different kernels.

Popular because

- Interactive Coding
- Combines codes & explanation
- Visual outputs
- Widely used in data science tools
- Great for learning & experimentation

Question 28

Python is a high level, general-purpose programming language used to write instructions that computers can understand & execute.

Popular because

- Easy to learn & read
- Powerfull libraries & tools
- Strong in data handling
- Widely used in AI & machine learning

Libraries

1) Pandas
- Used for data manipulation & analysis
- Helps work with tables
- Make it easy to clean & organize data

2) NumPy
- Used for numerical computing
- Works with arrays
- Foundation for many other data tools

3) Matplotlib
- Used for data visualization
- Creates charts, graphs, plots
- Helps make data easier to understand

4) Scikit-learn
- Used for machine learning
- Provides tools for building predictive models

Question 29

The concept of protecting personal information and ensuring that individuals have control over how their data is collected, stored, shared & used.

Importance

- Protects personal identity
- Builds Trust
- Prevents misuse of data
- Ensures legal Compliance

GDPR - a data protection law created by European Union to give individuals more control over their personal data & regulate how organizations collect, store & process data

Principles

- Consent
- Transparency
- Data Minimization
- Right to access
- Right to deletion
- Security

Countries

- EU countries (Germany, France, Italy, Spain
- Norway
- Iceland
- Liechtenstein

Question 30

Refers to the principles & guidelines that govern how data is collected, used, stored & shared in a responsible, fair & transparent way.

Unethical Use

1) Using personal data without consent
- Violates privacy rights
- Users don't know how their data is being used.

2) Biased AI decision-making
- Reinforces discrimination
- Produces inaccurate or unjust outcomes

3) Deepfakes & misinformation
- Spreads false information
- Damages reputations

Question 31

The set of policies, processes, roles & standards that ensure data is managed properly, securely, used responsibly within an organization

Components

- Data quality management
  - Ensures data is accurate, complete & reliable
  - Involves cleaning & validating data

- Data Policies & Standards
  - Ensures consistency across the organization

- Roles & Responsibilities
  - Ensures accountability
  - Defines who is responsible for data

- Data security & privacy
  - Protects data from unauthorized access or breaches

- Data access management
  - Uses permissions & authentication
    Controls who can view or use specific data

- Data lifecycle management
  - Ensures outdated unneccesary data is removed

- Monitoring & Compliance
  - Identifies & fixes issues early